

## HAY, Australia

WMO: 947020

Lat: 34.5411S

Lon: 144.8344E

Elev: 93

StdP: 100.22

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |            | Humidification DP/MCDB and HR |            |             |             |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|------------|-------------------------------|------------|-------------|-------------|------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |             |            | 99.6%                         |            |             | 99%         |            |             | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%       | 99%        | DP                            | HR         | MCDB        | DP          | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>7      | (b)<br>-0.3 | (c)<br>1.0 | (d)<br>-7.0                   | (e)<br>2.1 | (f)<br>23.9 | (g)<br>-5.1 | (h)<br>2.5 | (i)<br>22.4 | (j)<br>11.8           | (k)<br>13.6 | (l)<br>10.9 | (m)<br>14.1 | (n)<br>1.9            | (o)<br>250 | (p)<br>0.550 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>1      | (b)<br>15.3            | (c)<br>40.1     | (d)<br>19.6 | (e)<br>37.8 | (f)<br>19.0 | (g)<br>35.8 | (h)<br>18.6 | (i)<br>23.2         | (j)<br>30.3 | (k)<br>21.8 | (l)<br>30.0 | (m)<br>20.8 | (n)<br>29.5 | (o)<br>6.1           | (p)<br>340 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 21.7                            | 16.6 | 24.1 | 19.8 | 14.7 | 23.8 | 18.2 | 13.3 | 23.4 | 69.3          | 30.9 | 64.1 | 30.0 | 60.4 | 29.3 | 36.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 11.3              | 10.1 | 9.0 | -2.6                       | 43.2 | 1.5                | 2.5 | -3.6                                               | 44.9 | -4.5       | 46.4 | -5.3       | 47.8 | -6.4       | 49.5 |
|                   |      |     | -2.9                       | 26.2 | 1.3                | 5.0 | -3.8                                               | 29.8 | -4.6       | 32.7 | -5.3       | 35.5 | -6.3       | 39.1 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 17.2   | 25.7  | 25.0  | 21.6  | 17.0  | 12.9  | 10.1  | 9.4   | 10.5  | 13.4  | 17.0  | 20.7  | 23.3 |
|                                                                                  | DBStd       | 6.86   | 4.51  | 4.34  | 4.27  | 3.74  | 3.02  | 2.58  | 2.39  | 2.80  | 3.64  | 4.23  | 4.74  | 4.56 |
|                                                                                  | HDD10.0     | 108    | 0     | 0     | 0     | 0     | 6     | 30    | 41    | 26    | 6     | 1     | 0     | 0    |
|                                                                                  | HDD18.3     | 1290   | 1     | 3     | 15    | 69    | 169   | 247   | 276   | 243   | 155   | 79    | 25    | 7    |
|                                                                                  | CDD10.0     | 2729   | 488   | 419   | 359   | 209   | 97    | 32    | 23    | 41    | 108   | 217   | 321   | 414  |
|                                                                                  | CDD18.3     | 868    | 230   | 188   | 116   | 29    | 2     | 0     | 0     | 1     | 8     | 38    | 96    | 162  |
|                                                                                  | CDH23.3     | 11781  | 3231  | 2406  | 1379  | 340   | 18    | 0     | 0     | 10    | 136   | 591   | 1353  | 2317 |
|                                                                                  | CDH26.7     | 6175   | 1902  | 1318  | 642   | 100   | 1     | 0     | 0     | 1     | 35    | 237   | 686   | 1253 |
| Wind                                                                             | WSAvg       | 4.5    | 4.9   | 4.8   | 4.5   | 3.9   | 4.0   | 3.8   | 4.3   | 4.5   | 4.8   | 4.9   | 5.0   | 5.1  |
| Precipitation                                                                    | PrecAvg     | 398    | 38    | 28    | 32    | 29    | 40    | 30    | 32    | 34    | 38    | 38    | 24    | 30   |
|                                                                                  | PrecMax     | 837    | 191   | 204   | 182   | 151   | 134   | 74    | 92    | 70    | 100   | 150   | 143   | 115  |
|                                                                                  | PrecMin     | 165    | 0     | 0     | 0     | 1     | 1     | 5     | 3     | 1     | 4     | 0     | 0     | 0    |
|                                                                                  | PrecStd     | 148    | 41    | 42    | 42    | 30    | 30    | 20    | 23    | 19    | 25    | 34    | 24    | 30   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 43.4  | 42.1  | 38.5  | 32.3  | 25.6  | 20.8  | 20.4  | 24.7  | 30.2  | 35.3  | 39.3  | 41.1 |
|                                                                                  |             | MCWB   | 19.9  | 19.9  | 19.3  | 16.7  | 14.7  | 13.7  | 12.0  | 12.8  | 14.7  | 15.9  | 18.4  | 18.6 |
|                                                                                  | 2%          | DB     | 40.7  | 38.6  | 35.1  | 29.4  | 23.0  | 18.4  | 17.7  | 21.2  | 27.0  | 32.1  | 36.1  | 38.3 |
|                                                                                  |             | MCWB   | 19.8  | 20.0  | 18.5  | 16.0  | 14.0  | 12.7  | 11.3  | 12.0  | 14.2  | 15.6  | 17.9  | 18.4 |
|                                                                                  | 5%          | DB     | 38.1  | 36.1  | 32.6  | 27.3  | 20.8  | 16.7  | 15.9  | 18.7  | 24.3  | 29.2  | 33.5  | 35.9 |
|                                                                                  |             | MCWB   | 19.3  | 19.5  | 18.1  | 15.6  | 13.6  | 11.8  | 10.9  | 11.1  | 13.6  | 15.1  | 17.3  | 18.2 |
|                                                                                  | 10%         | DB     | 35.5  | 33.9  | 30.3  | 24.9  | 18.9  | 15.2  | 14.5  | 16.7  | 21.5  | 26.6  | 30.7  | 33.4 |
|                                                                                  |             | MCWB   | 18.8  | 18.9  | 17.3  | 15.0  | 13.0  | 11.3  | 10.3  | 10.5  | 12.6  | 14.3  | 16.1  | 17.4 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 24.7  | 25.0  | 22.4  | 19.2  | 17.5  | 15.2  | 14.8  | 15.0  | 17.2  | 19.5  | 25.3  | 30.6 |
|                                                                                  |             | MCDB   | 29.4  | 29.2  | 28.7  | 25.1  | 19.5  | 18.3  | 16.5  | 20.3  | 23.4  | 26.7  | 25.3  | 30.9 |
|                                                                                  | 2%          | WB     | 22.4  | 23.0  | 21.1  | 18.0  | 15.6  | 13.7  | 12.9  | 13.4  | 15.9  | 17.7  | 20.1  | 21.7 |
|                                                                                  |             | MCDB   | 31.6  | 31.0  | 27.7  | 23.5  | 19.9  | 16.7  | 15.4  | 18.2  | 22.6  | 25.3  | 28.7  | 27.5 |
|                                                                                  | 5%          | WB     | 21.4  | 21.9  | 19.9  | 17.0  | 14.6  | 12.8  | 11.6  | 12.2  | 14.7  | 16.5  | 18.9  | 20.1 |
|                                                                                  |             | MCDB   | 31.1  | 30.1  | 27.7  | 23.4  | 18.9  | 15.5  | 14.7  | 16.9  | 21.3  | 25.2  | 27.6  | 28.8 |
|                                                                                  | 10%         | WB     | 20.5  | 20.8  | 18.7  | 15.9  | 13.7  | 11.9  | 10.7  | 11.2  | 13.4  | 15.4  | 17.8  | 18.8 |
|                                                                                  |             | MCDB   | 31.0  | 29.4  | 27.3  | 23.7  | 18.1  | 14.6  | 13.8  | 15.7  | 20.4  | 25.0  | 27.0  | 30.2 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 15.3  | 14.2  | 13.8  | 13.7  | 11.9  | 10.6  | 10.6  | 12.3  | 13.9  | 15.1  | 14.9  | 15.2 |
|                                                                                  |             | MCDBR  | 17.8  | 16.5  | 15.6  | 16.0  | 13.8  | 12.6  | 12.4  | 15.4  | 17.2  | 18.2  | 17.5  | 17.1 |
|                                                                                  |             | MCWBR  | 5.8   | 5.7   | 5.8   | 7.0   | 7.0   | 7.6   | 7.4   | 8.1   | 7.4   | 7.1   | 6.1   | 6.2  |
|                                                                                  | 5% WB       | MCDBR  | 13.8  | 12.6  | 12.1  | 12.9  | 10.7  | 10.1  | 10.5  | 12.7  | 14.4  | 14.7  | 13.7  | 13.2 |
|                                                                                  |             | MCWBR  | 6.2   | 5.7   | 5.4   | 6.6   | 6.1   | 6.5   | 6.8   | 7.5   | 7.3   | 7.1   | 6.0   | 7.0  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.336  | 0.338 | 0.317 | 0.305 | 0.295 | 0.288 | 0.288 | 0.295 | 0.325 | 0.327 | 0.342 | 0.333 |      |
|                                                                                  | taud        | 2.490  | 2.503 | 2.550 | 2.551 | 2.553 | 2.573 | 2.567 | 2.536 | 2.442 | 2.458 | 2.448 | 2.482 |      |
|                                                                                  | Ebn at Noon | 1001   | 977   | 962   | 918   | 873   | 854   | 873   | 915   | 937   | 977   | 987   | 1007  |      |
|                                                                                  | Edh at Noon | 115    | 109   | 97    | 86    | 75    | 69    | 73    | 85    | 106   | 113   | 119   | 117   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 7.88   | 7.01  | 5.88  | 4.35  | 3.12  | 2.50  | 2.73  | 3.71  | 5.10  | 6.45  | 7.22  | 7.96  |      |
|                                                                                  | RadStd      | 0.42   | 0.35  | 0.32  | 0.19  | 0.15  | 0.17  | 0.15  | 0.23  | 0.32  | 0.38  | 0.41  | 0.34  |      |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (8 neighbors) | +0.43 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +157    | +126    |

Nomenclature: See separate page